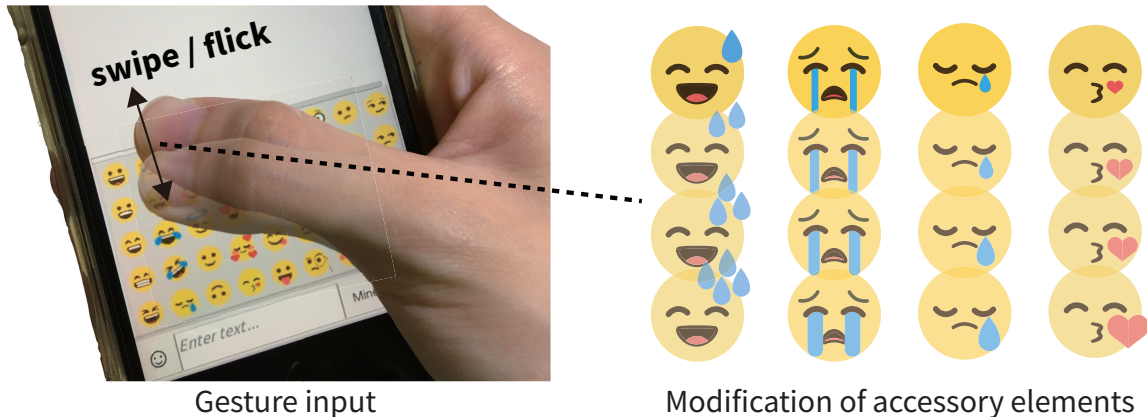


# On-the-fly Editing of Emoji Elements for Mobile Messaging

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**Figure 1:** Users can rapidly edit facial emojis by editing the number and scale of accessory elements via swiping and flicking when typing a text message on a mobile.

## ABSTRACT

This poster presents an on-the-fly emoji editing method to extend the range of emotional expressions of off-the-shelf emojis when typing a message on a mobile. The proposed system allows for the quick editing of accessory elements of facial emojis via simple swipe or flick gestures to reflect the user's emotional intensity.

## CCS CONCEPTS

• Human-centered computing → Interaction techniques.

## KEYWORDS

emoji, on-the-fly editing, mobile messaging

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## 1 INTRODUCTION

Emojis are widely used in text communication, especially messaging applications, to express difficult-to-describe information such

as emotions and feelings. Users can convey diverse and subtle expressions using many emojis that have been added to the Unicode standard. Commercial software also provides application-specific emojis, such as animating and sound-producing emojis. Accordingly, several input interfaces have been developed to efficiently manage the large emoji dataset via zooming in/out interaction with a single panel [5] and gesture command interface [1].

User customization of an off-the-shelf emoji is another direction to increase their variety. For example, Apple memoji allows the creation of customized facial emojis based on the geometric features and expression of a motion-captured user's face. A sketch-based method [4] has been proposed for creating personalized emojis using generative adversarial networks. Some chat tools, such as Slack, can import a hand-crafted image as an emoji. However, users must register such customized emoji to the keyboard before messaging. Hence, communication should be interrupted if new emojis are needed when sending a message. To overcome this problem, on-the-fly customization methods have been proposed to edit emojis while typing messages. For instance, MojiBoard [1] scales the emoji according to the size of gesture inputs. Google's Gboard synthesizes an intermediate emoji by combining two emojis. VibEmoji [2] allows users to compose multi-modal emojis by adding animation and vibration effects.

Our motivation for on-the-fly editing of off-the-shelf emojis is to expand the range of emotional expressions when typing a message on a mobile device. We assume that emotional expressions can be exaggerated by editing the accessory elements drawn on the face, such as tears 😓, sweat 💦, and hearts 🥰<sup>1</sup>, which play a particularly significant role in emotional expressions [3]. For example, a sad

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<sup>1</sup>Twemoji by Twitter, Inc and other contributors is licensed under CC-BY 4.0

tearful face 😭 can be exaggerated by changing the amount of tears. We have assumed three requirements to achieve quick emoji editing in mobile messaging.

- Simplicity and operation speed are more important than emoji shape customizability because prompt communication is often required in text messaging. Quick editing is also crucial to reflect the user's emotions without delay.
- Emoji shape should be continuously controlled according to emotional intensity. At least, shape control in multiple discrete steps should be possible.
- Maximum compatibility with the standard keyboard should be maintained to minimize user learning costs.

Under these assumptions, our proposed system defines only one controllable parameter for each facial emoji to edit the accessory elements quickly. The shape parameter controls the scale or the number of accessories as shown in Figure 1. Our emoji keyboard provides a two-stage input function; the user selects a facial emoji on a list or keyboard and specifies the shape parameter value if necessary. This editing strategy ensures fine control of emotion and intensity in facial emojis while maintaining compatibility with the standard keyboard interface.

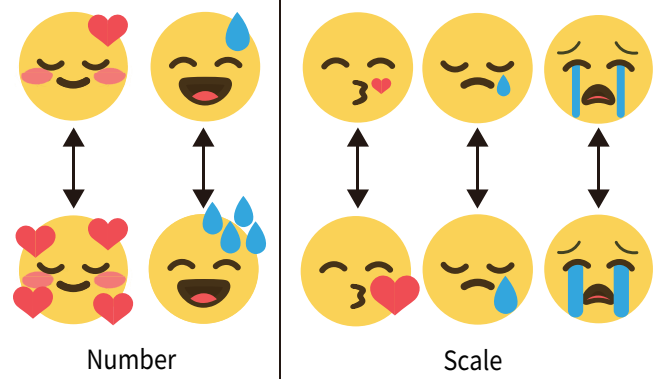
## 2 PROPOSED SYSTEM

### 2.1 Customizable Emojis

Our system parameterizes each facial emoji with a single parameter to change their emotional impression as shown in Figure 2. We empirically define different shape parameters for each emoji to control the scale or number of accessories. For example, the prototype system controls the smiling emoji with the number of sweats 💦 and hearts 🥰. The number of accessories is discretely varied between one and four based on the continuous user input. The element placement is manually predetermined to preserve the original graphical design. Conversely, the emotional impression of 😭😭😭 is emphasized by scaling the heart and falling tears. Our system does not change the number and placements of these accessories as they are arranged in relation to the locations of facial parts such as eyes and mouth.

### 2.2 On-the-fly Emoji Editing

The user operation on our emoji interface is composed of two stages. First, the user selects a facial emoji from the standard list or keyboard. Next, the shape parameter is changed continuously by gesture input if necessary. Among many methods to specify a continuous parameter on mobile devices, we examined two types of sliding gesture-based methods as the first step of our research. One method determines the shape parameter value according to the velocity of the flicking gesture on the emoji. This method facilitates instant operation, whereas several trials and errors are required to achieve the ideal shape as the edited result should be checked after every flicking operation. The other is a swipe-based method that computes the shape parameter according to the swipe distance from the emoji location. This approach allows the user to tweak the shape parameter with realtime editing output. However, it is less effective than the flick-based method, especially for specifying



**Figure 2: Each facial emoji is edited with a single shape parameter associated with the number and scale of the accessories.**

discrete values, such as the number of accessories. We will explore more efficient and intuitive ways to manipulate accessory shapes.

## 3 DISCUSSION

We performed a preliminary experiment to validate the proposed method. The experimental results indicated that the message senders and receivers quickly recognized the change in the number of accessories. Conversely, differences in accessory size were intuitively changed by the sender, whereas the receivers were less aware of relative size differences because the reference size was not available. We could apply extreme scaling for enhanced perception, but it damages the original design. One possible solution might be controlling multiple shape features simultaneously with user control. Our future work also includes the development of an on-the-fly animation control method for more expressive emoji communication. For example, we assume that the intensity of a sad expression can be exaggerated by temporally changing the speed and amount of tear flow.

In summary, we have presented a simple editing method for accessories of facial emojis to represent the intensity of the sender's emotion. Our approach might contribute to alleviating the increase of the emoji set; a slight emotional difference could be expressed by modifying existing emojis without creating a new one.

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